WEEK 2 · SATURDAY

Calculus for AI

How a machine finds out which way is downhill.

AI for Beginners · Lesson 2, Part A



TODAY'S JOURNEY

From speedometers to AI

1. **Why calculus?**
   AI learns by small nudges. Calculus picks the nudge.
2. **Rate of change**
   Average speed versus the speedometer.
3. **The derivative**
   The slope at one exact point.
4. **A few handy rules**
   Power, sum and chain. That is all we need.
5. **Many knobs**
   Partial derivatives: change one thing at a time.
6. **The gradient**
   An arrow pointing uphill. We walk the other way.

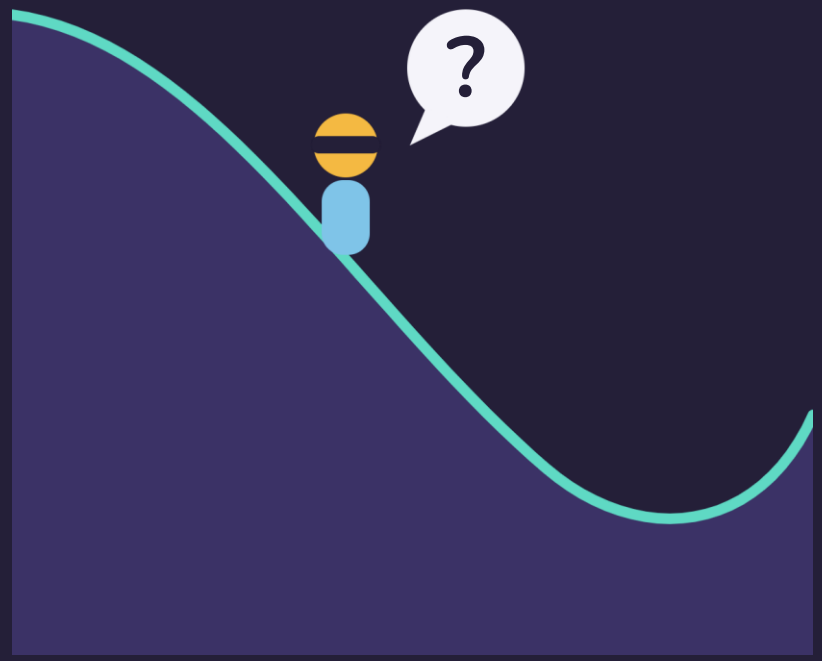
No heavy algebra today. We want the idea, because you will meet it again in every AI lesson that follows.

WHERE WE LEFT OFF

A network learns by rolling downhill.

But how does it know which way is down?

That is the question calculus answers.



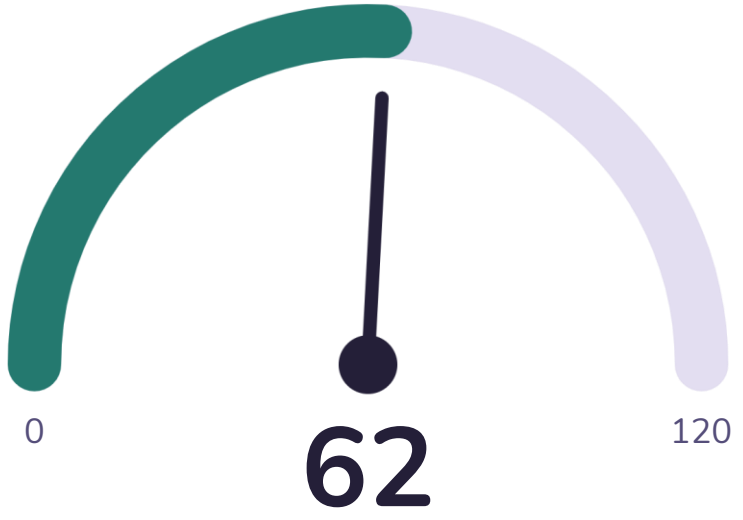
RATE OF CHANGE

Average speed vs speedometer

You drive **100 km in 2 hours**. Your **average speed** is 50 km/h.

But at minute 43, was the needle on exactly 50? **No.** Traffic lights, hills, open road.

> **Key idea**
>
> **Average rate** is the big picture over a stretch. **Instant rate** is what is happening right now.



RATE OF CHANGE

Average rate is a slope between two points

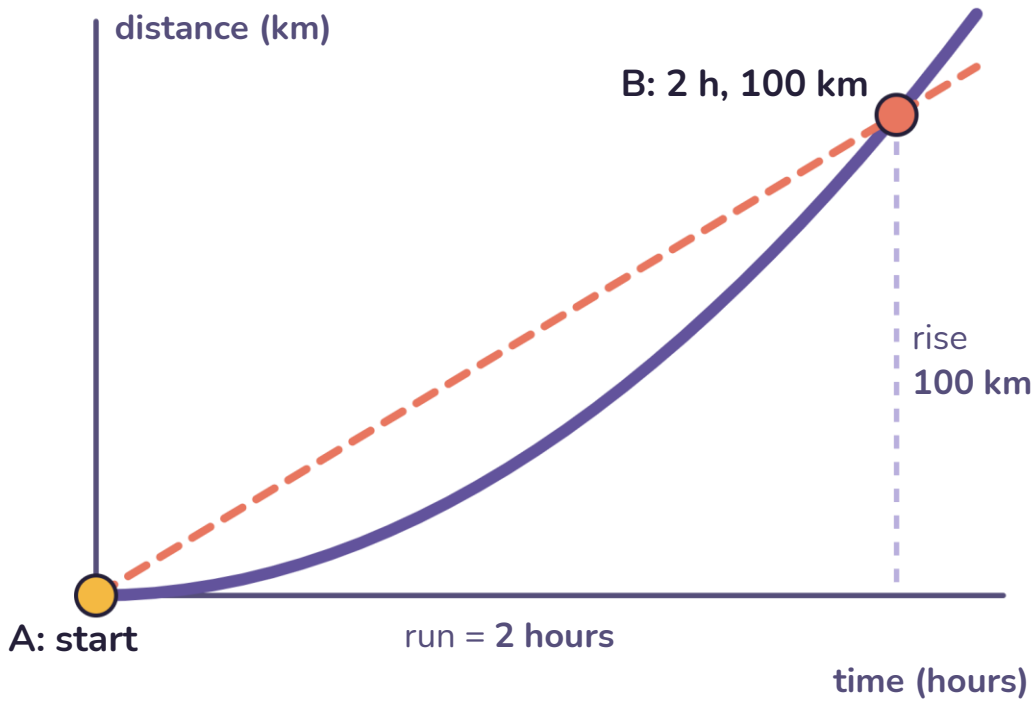
Pick two moments on the trip and draw a line through them.

A to B: 100 km in 2 hours

slope = rise ÷ run

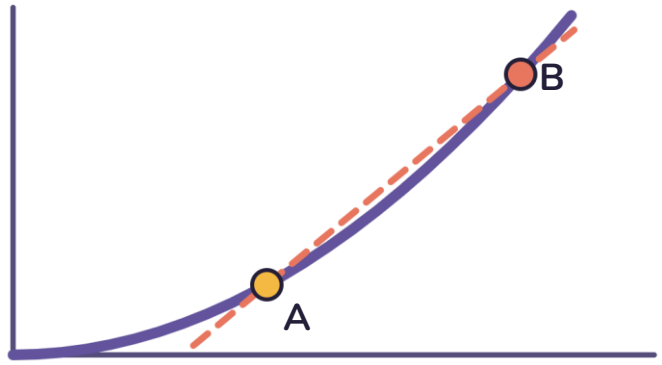
= 100 ÷ 2 = 50 km/h

The slope of that line is the **average rate of change**. Its maths name is a **secant line**.



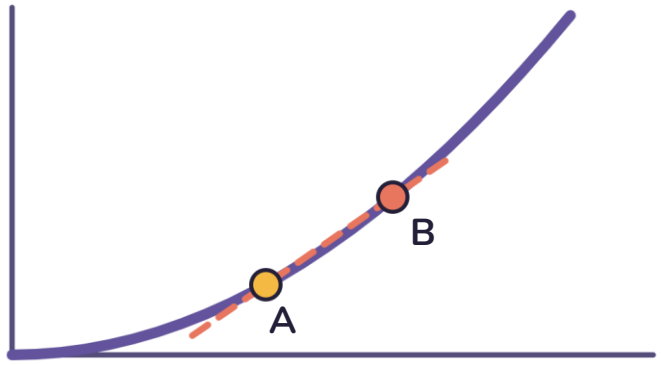
FROM AVERAGE TO INSTANT

Zoom in: squeeze B toward A



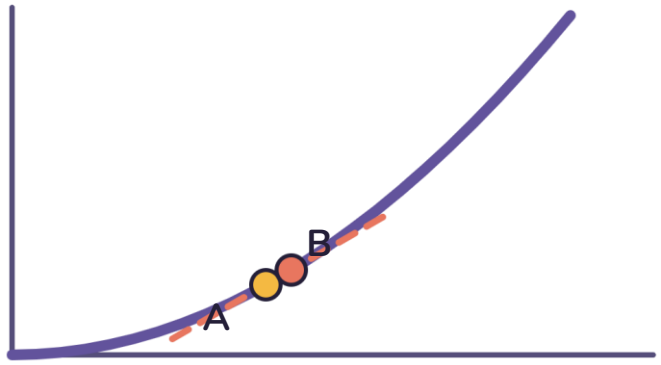
B is **1 hour** away

slope = 75



B is **0.5 hour** away

slope = 62.5



B is **0.1 hour** away

slope = 52.5

Keep squeezing. The slope settles on **50 km/h**, the speed at exactly 1 hour. That limit is the **derivative**.

THE DERIVATIVE

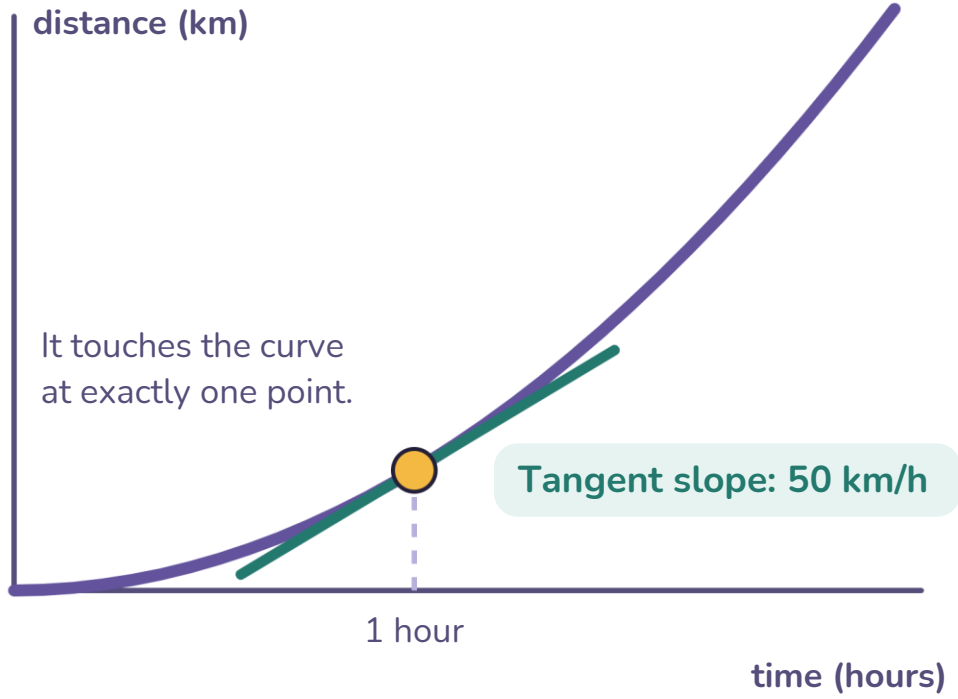
The slope at one exact point

$$f'(x) = [\, f(x + h) - f(x) \,] \div h$$

...as h shrinks toward zero

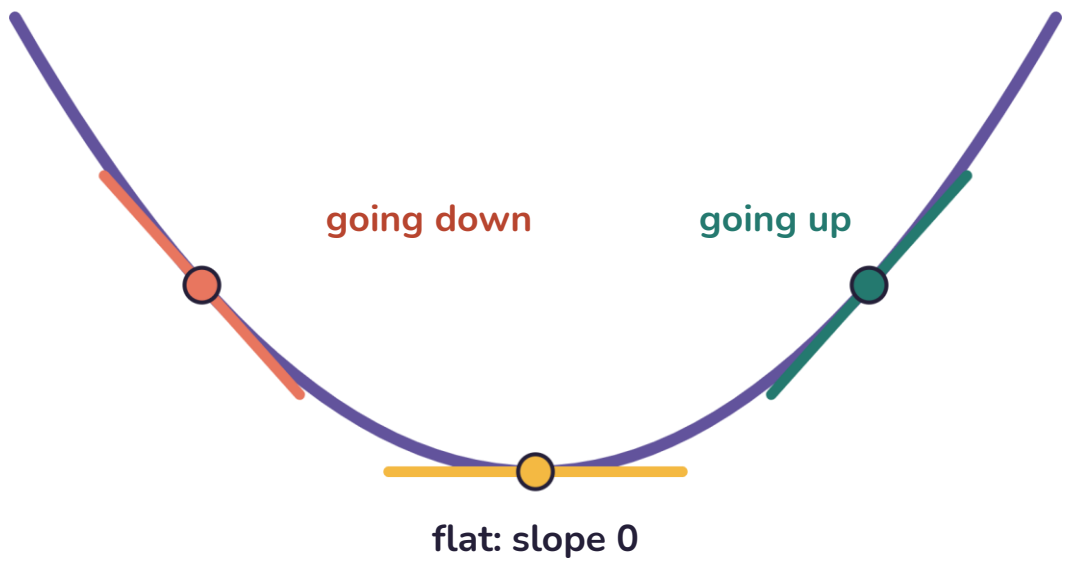
1. Nudge the input by a tiny amount, h.
2. Measure how much the output changed.
3. Divide: change in output ÷ change in input.

In AI: f is the error, x is a setting. f' says how the error reacts when we nudge that setting.



THE DERIVATIVE

The sign of the slope tells the story



f'(x) < 0

Nudge x up and f goes down.

f'(x) = 0

Flat. A valley or a peak. Maybe the answer!

f'(x) > 0

Nudge x up and f goes up.

THE TOOLKIT

Four rules are enough to start

Rule	Example	Derivative	In words
Constant	5	0	A flat line never changes.
Power	x^3	$3x^2$	Bring the power down, lower it by 1.
Constant multiple	$4x^3$	$12x^2$	Numbers in front tag along.
Sum	$x^2 + 3x$	$2x + 3$	Do each piece, then add.

Power rule in one line

x^n becomes n · x^(n−1). So for x^3 at x = 2, the slope is $3 \cdot 2^2 = \mathbf{12}$.

WORKED EXAMPLE

Differentiate it piece by piece

Find the derivative of $f(x) = 4x^3 - 5x^2 + 7x - 9$

PIECE 1

$4x^3 \rightarrow 12x^2$

Power rule: $4 \cdot 3 = 12$, power drops to 2.

PIECE 2

$-5x^2 \rightarrow -10x$

Power rule: $-5 \cdot 2 = -10$, power drops to 1.

PIECE 3

$7x \rightarrow 7$

A plain x has slope 1, so $7 \cdot 1 = 7$.

PIECE 4

$-9 \rightarrow 0$

Constants never change.

$$f'(x) = 12x^2 - 10x + 7$$

Plug in $x = 1$ and the slope is $12 - 10 + 7 = \mathbf{9}$. Positive, so f is still climbing there.

QUICK CHECK

Your loss curve has slope −4 at the current setting. You nudge the setting up a little. What happens to the loss?

A. It goes up

B. It goes down

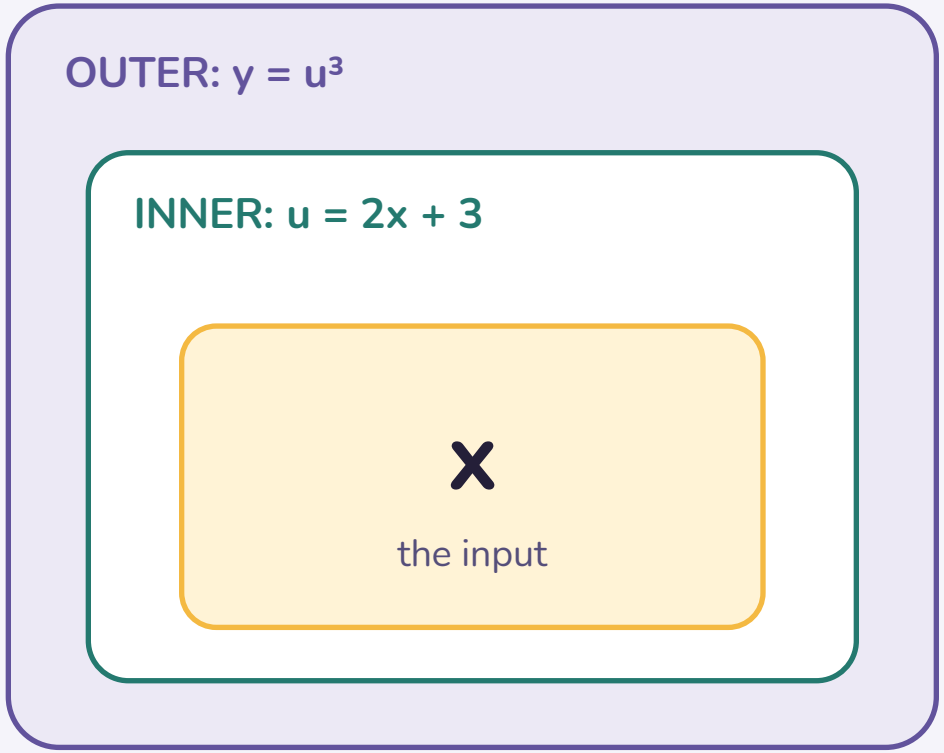
C. It stays the same

D. Cannot tell

Think for 30 seconds, then answer in the chat.

FUNCTIONS INSIDE FUNCTIONS

The chain rule: peel from the outside in



1. Name the inner part: **u = 2x + 3**. Now y = u³.
2. Outer slope: **dy/du = 3u²**
3. Inner slope: **du/dx = 2**
4. Multiply: dy/dx = 3u² · 2 = **6(2x + 3)²**. At x = 1 that is 6 · 25 = **150**.

LAYERS PASS CHANGES ALONG

Why AI loves the chain rule



Nudge x by 1. Each layer multiplies the change it receives:

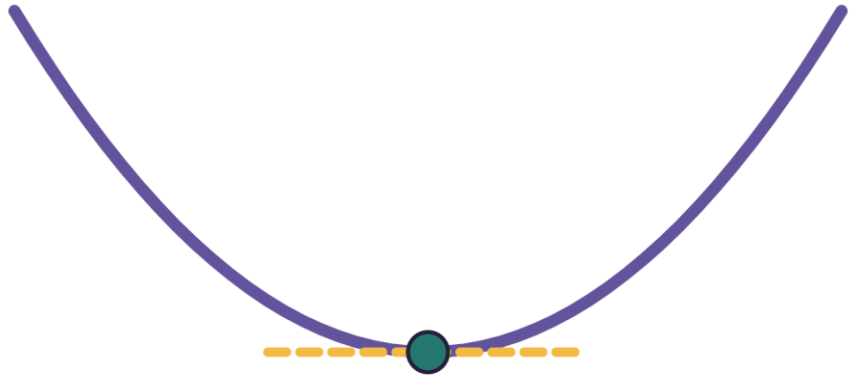
$$2 \times 3 \times 0.5 = 3$$

So the loss moves by 3. That is the chain rule: **multiply each layer's sensitivity**.

d loss/dx = d loss/dv · dv/du · du/dx — Backpropagation does this, layer by layer

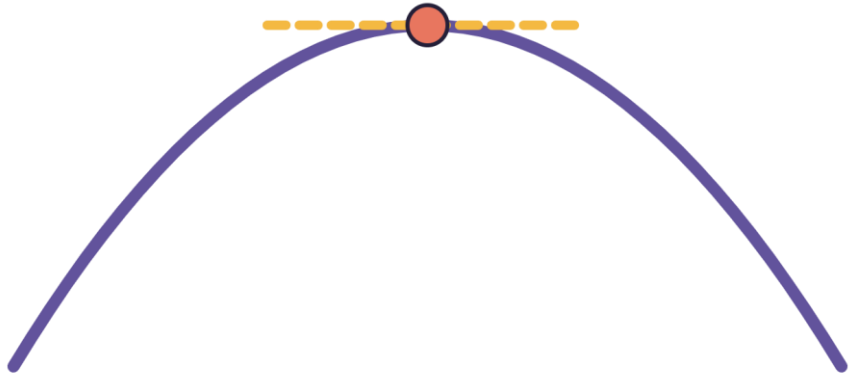
FINDING THE BOTTOM

Valley or hilltop? Ask the second derivative



f'(x) = 0 and f''(x) > 0

Bends like a cup: a MINIMUM



f'(x) = 0 and f''(x) < 0

Bends like a hill: a MAXIMUM

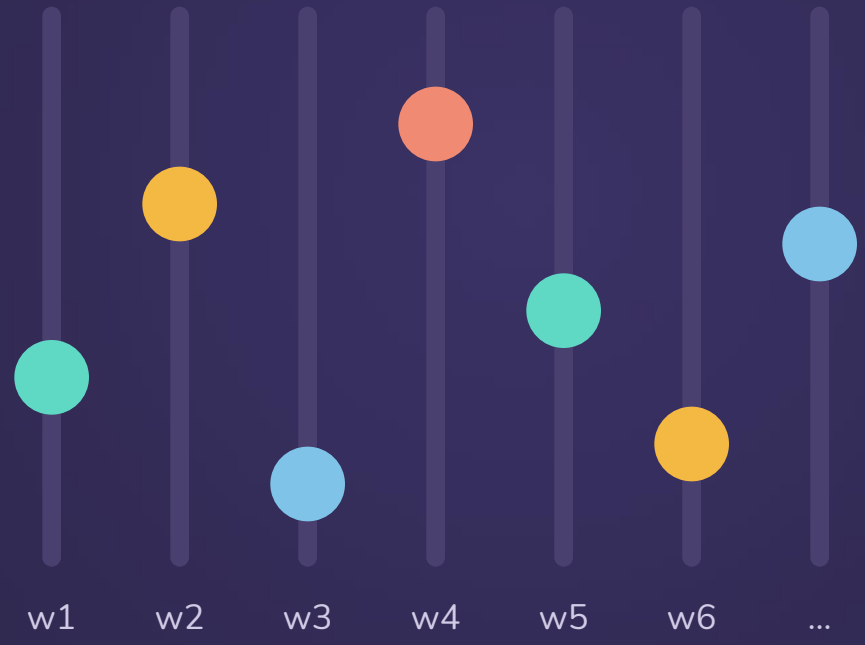
f'' is the slope of the slope. Training an AI means hunting for the valley of its loss curve.

THE PROBLEM

Real models have millions of knobs

Chatbot-sized models have billions of settings, called **weights**. One input variable is not enough. We must handle many at once.

Enter: partial derivatives.

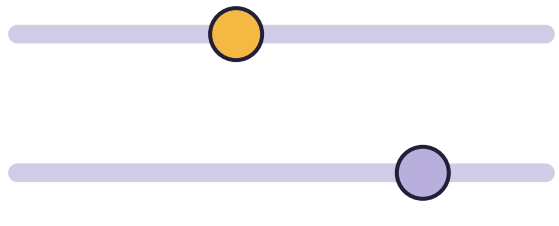


MANY INPUTS

Partial derivatives: one knob at a time

Think of baking a cake

The score depends on sugar and oven time. Change **only the sugar**, keep the oven fixed, and watch the score. That is **∂score/∂sugar**.



The rule

The curly **∂** means: change just this one input and treat the others as plain numbers.

$$f(w1, w2) = 3w1^2 + 4w1w2 + w2^3$$

$\partial f/\partial w1 = \mathbf{6w1 + 4w2}$

$\partial f/\partial w2 = \mathbf{4w1 + 3w2^2}$

At w1 = 1 and w2 = 2, the two slopes are **14** and **16**.

MANY INPUTS

The gradient: every slope in one arrow

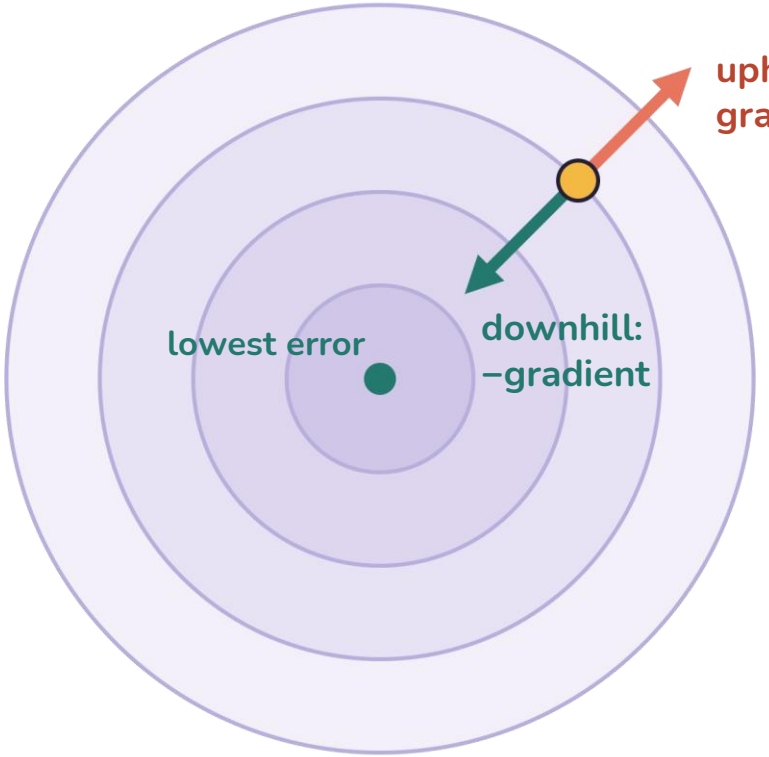
Collect the partial derivative of every knob into one list. That list is the **gradient**.

$$\nabla f = [\ \partial f/\partial w1\ ,\ \partial f/\partial w2\ ]$$

Our example at (1, 2): $\nabla f = [14, 16]$

The whole point

The gradient points the **steepest way uphill**. To cut the error, walk the **opposite way**: $-\nabla f$.



RECAP

Three things to remember

The derivative

The slope at one exact point. How strongly the output reacts to a tiny nudge.

The sign of the slope

Positive means going up, negative means going down, zero means flat: maybe the bottom.

The gradient

One arrow made of every knob's slope. AI walks the opposite way to shrink its error.

Chain rule: layers pass changes along. Multiply each layer's sensitivity.

Tomorrow: loss functions, gradient descent in action, probability and matrices.

BEFORE SUNDAY

Try it yourself

1. Differentiate **f(x) = 3x² + 2x − 7**
2. Chain rule: find dy/dx for **y = (x + 1)^4**
3. Think: name one thing in your life that has a rate of change. Is it average or instant?

Bring your answers on Sunday. We start with them.